

ABSTRACT

The rise of micro-drones poses significant security and privacy threats. To address this challenge, we propose a hybrid deep learning model that leverages both audio and image data to enhance detection accuracy. The model extracts spectral-temporal features from acoustic signals and texture-shape features from visual imagery, which are then fused to capture cross-modal patterns indicative of drone activity. Several machine learning models, including logistic regression, SVM, random forest, InceptionV3, ResNet and DenseNet121, were trained and evaluated for their effectiveness. Our InceptionNet-based deep learning architecture achieves 100% detection accuracy, outperforming traditional machine learning techniques such as SVM, which achieved 97%. This multi-modal approach effectively minimises false alarms and improves detection reliability under challenging conditions.

INTRODUCTION

- Micro-drones (< 2kg) have become accessible and hence have more widespread adoption due to technological advancements [1].
- While they are significant across commercial, recreational and military applications, they have been used in malicious activities which include trafficking and spying [2].
- In response to these hostile drone uses, the development of anti-drone systems (ADS) is crucial, which have three key functionalities: detection, localization, and neutralization.
- The detection unit plays a crucial role in identifying the presence of a drone in restricted areas based on sensor data [3]. Hence, developing an efficient detection model is of prime importance for the development of ADS.

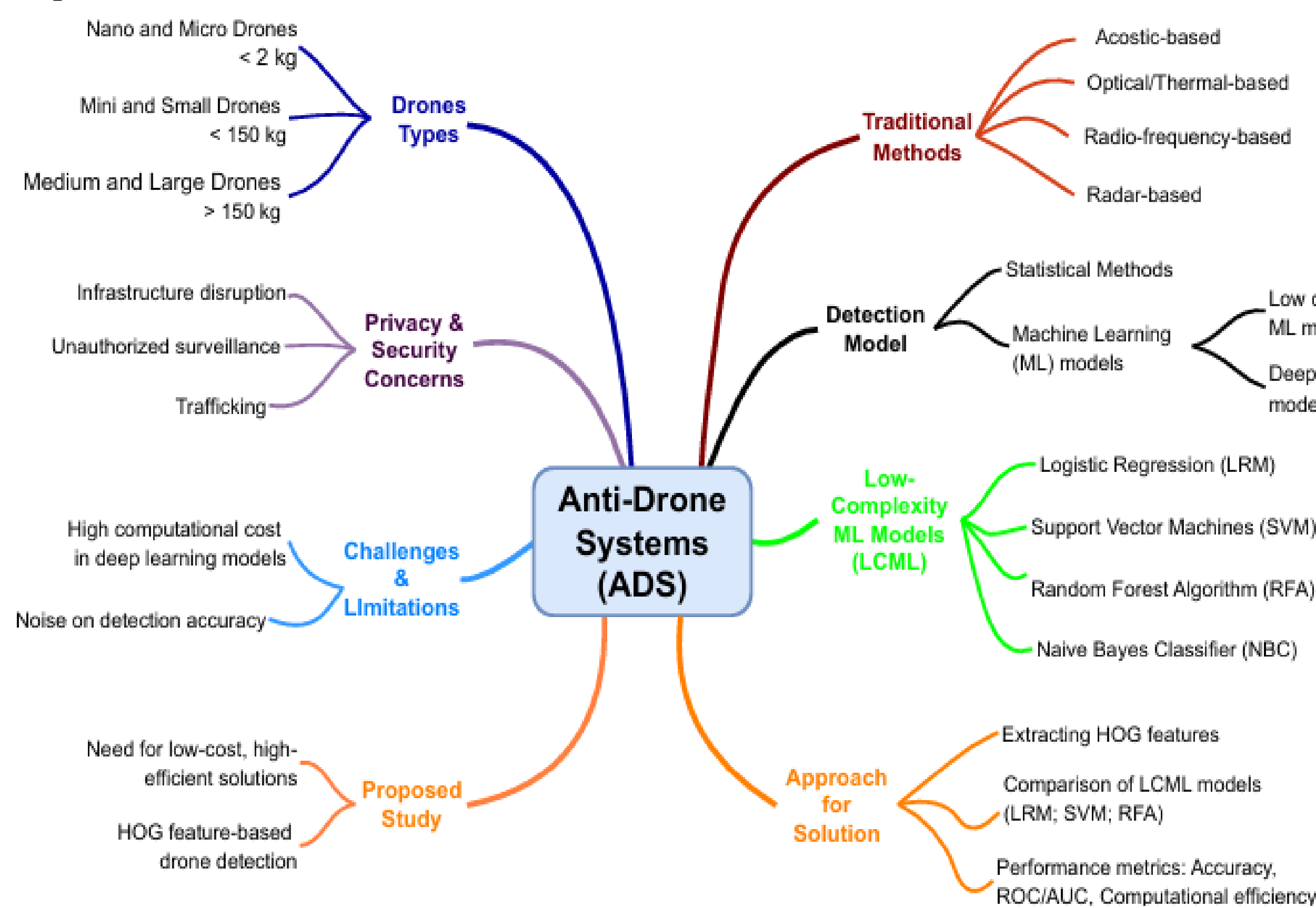


Figure 1. Mind map of Anti-Drone System components and detection methods.

Challenges of Traditional Drone Detection Methods

Traditional single modality methods typically face limitations such as restricted detection range, susceptibility to environmental noise, and inefficiency in detecting stealthy micro-drones [4].

Objective

Development of a deep learning model that can detect a micro-drone in a restricted area with high accuracy.

Significance

Enhancement of drone detection and identification capabilities. Therefore, contributing towards robust anti-drone systems to combat drone misuse.

METHODOLOGY

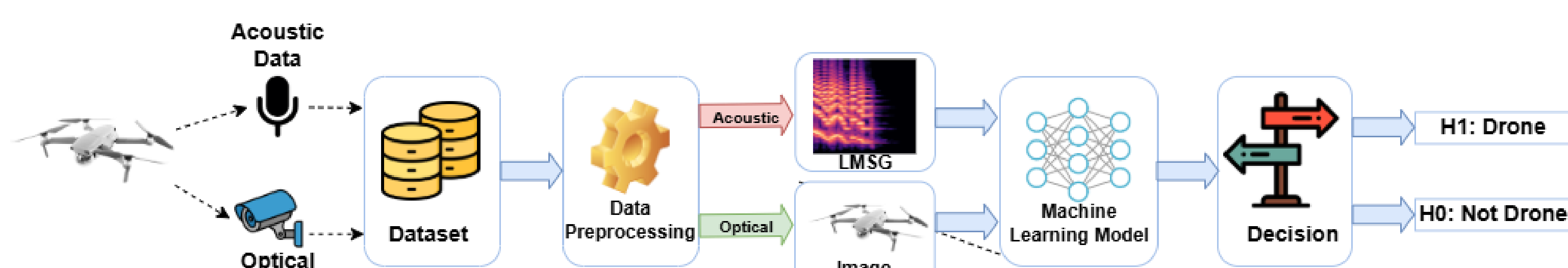


Figure 2. Block diagram of drone detection using acoustic and visual features.

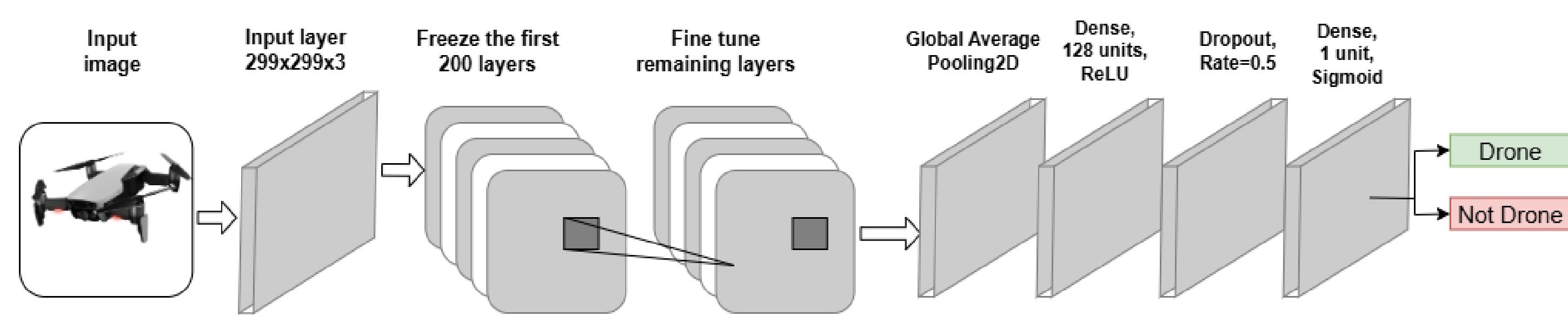


Figure 3. Architecture of InceptionV3 with custom tuning.

RESULTS

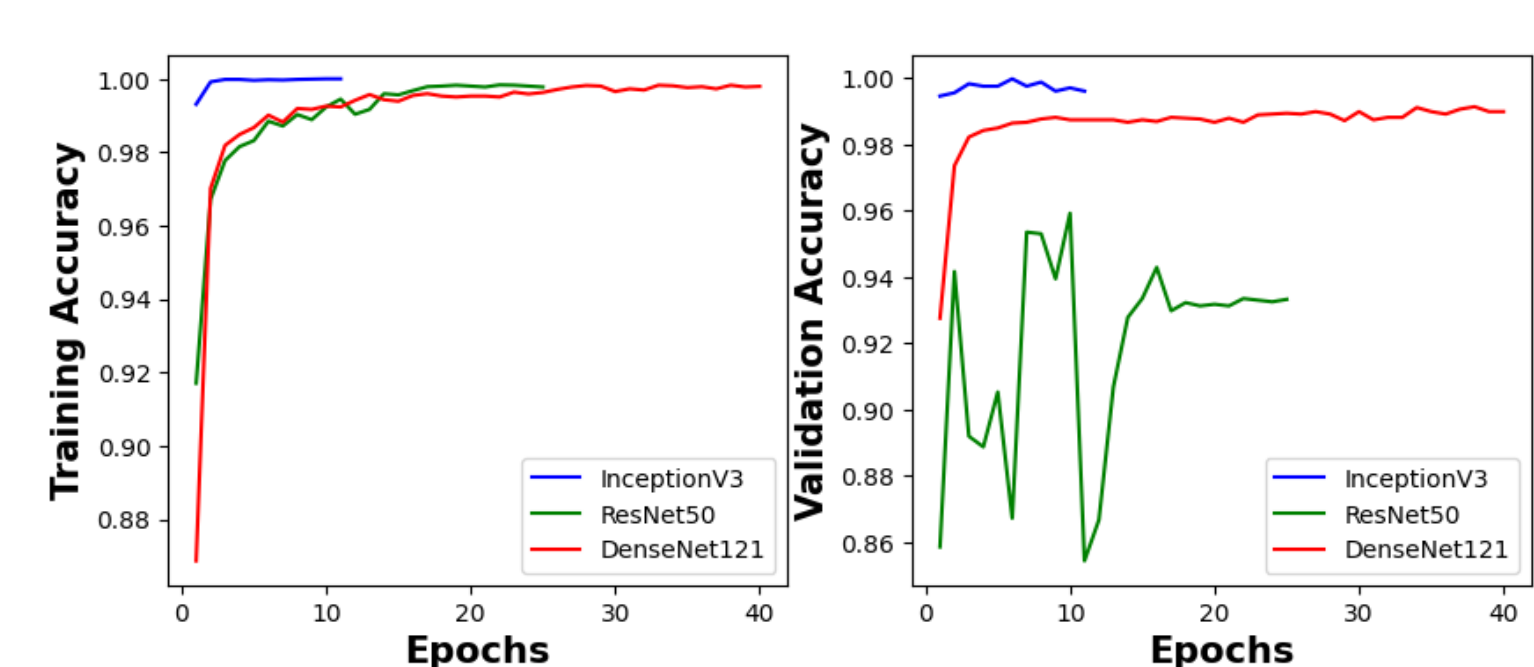


Figure 4. Training and Validation Accuracy vs Epoch Curves

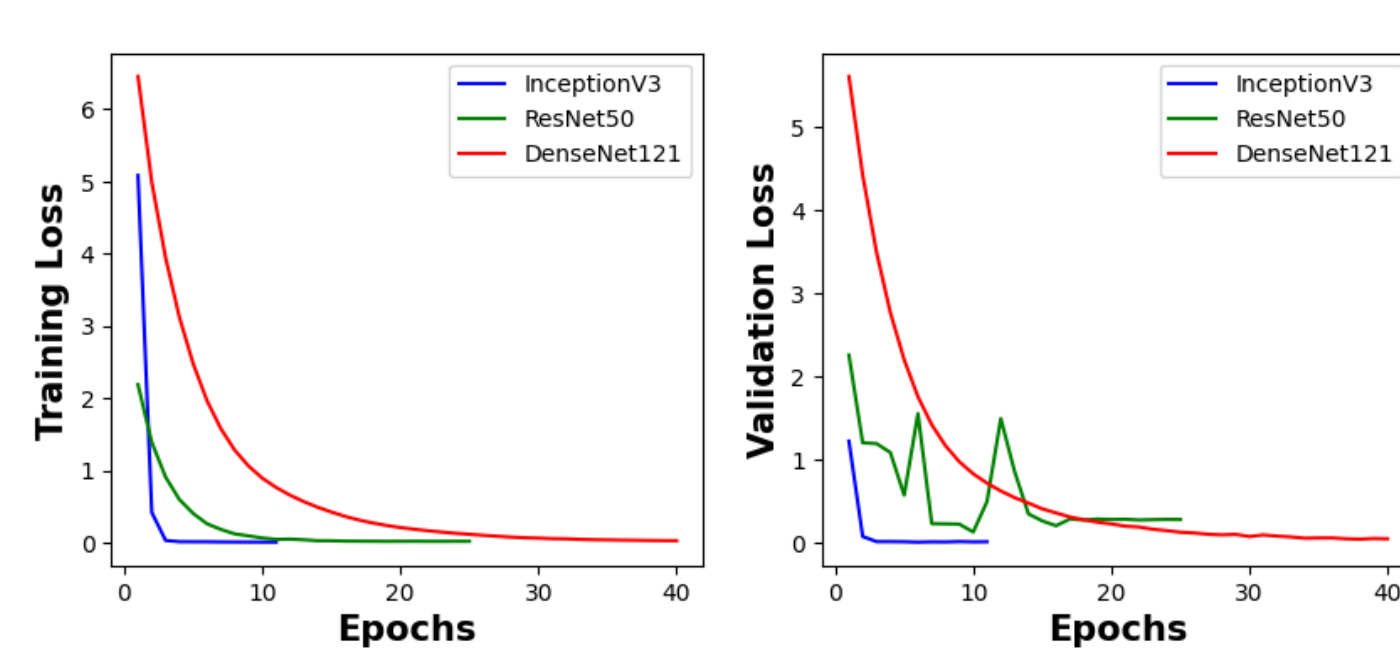


Figure 5. Training and Validation Loss vs Epoch Curves

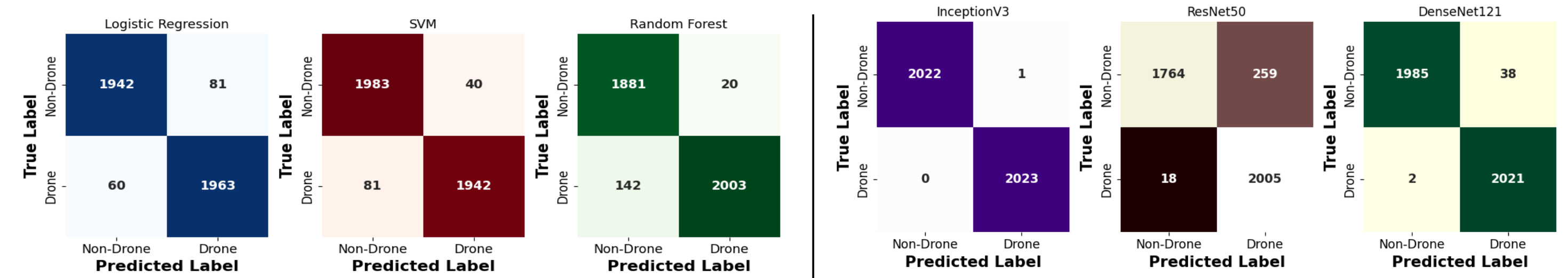


Figure 6. Confusion Matrices for Logistic Regression, SVM and Random Forest

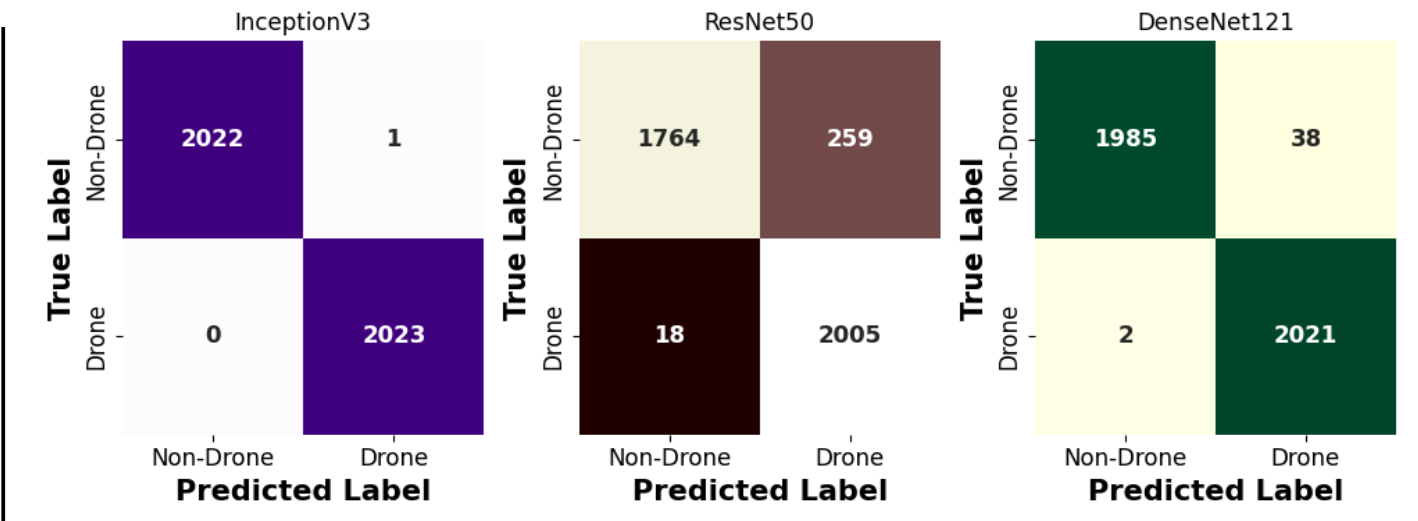


Figure 7. Confusion Matrices for InceptionV3, ResNet50 and DenseNet121

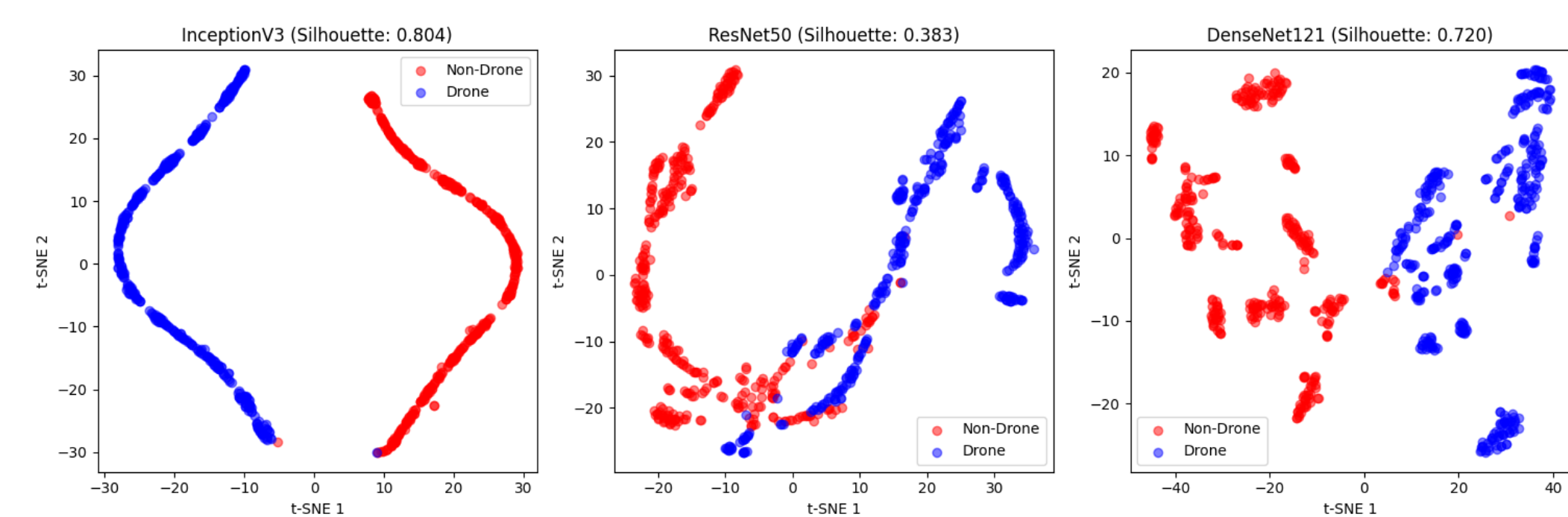


Figure 8. t-SNE plots for showing the separation of extracted features for deep learning models

Table 1: Model classification metrics

Metric	Class	Logistic Regression	SVM	Random Forest	Inception v3	ResNet5 0	DenseNet 121
Precision	Not-Drone	0.97	0.96	0.99	1.00	0.99	1.00
	Drone	0.96	0.95	0.94	1.00	0.89	0.98
Recall	Not-Drone	0.96	0.95	0.93	1.00	0.88	0.98
	Drone	0.97	0.96	0.99	1.00	0.99	1.00
F1-Score	Not-Drone	0.96	0.96	0.96	1.00	0.93	0.99
	Drone	0.96	0.96	0.96	1.00	0.94	0.99
Accuracy		0.95	0.97	0.96	1.00	0.93	0.99

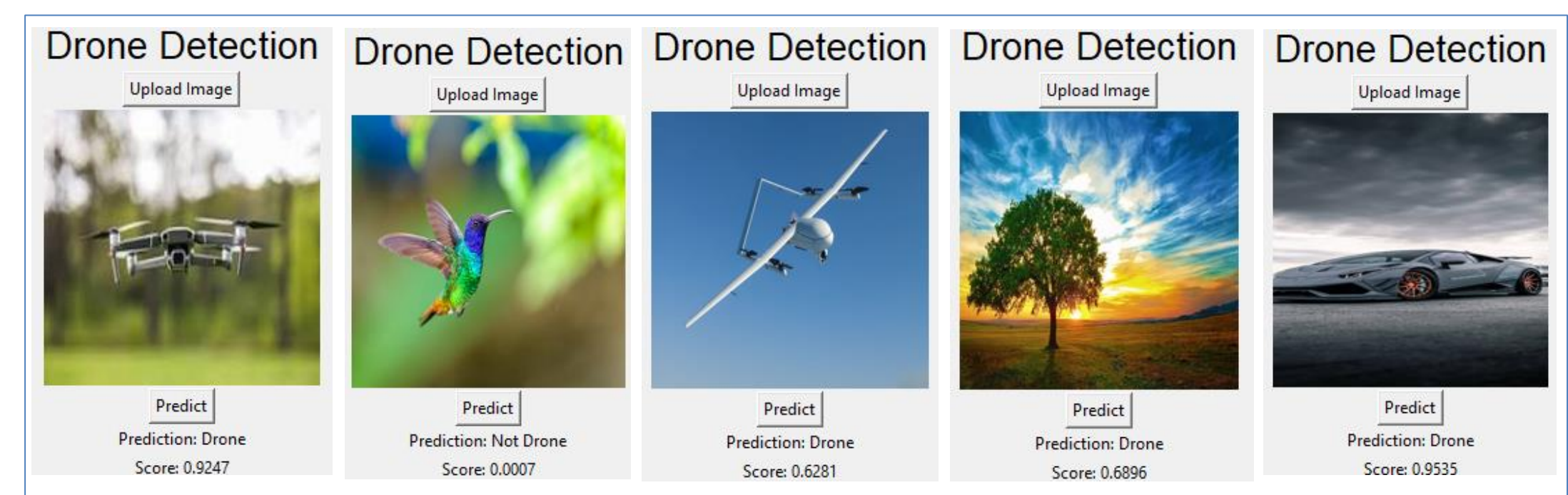


Figure 9. Inference Testing

CONCLUSION

- The study leveraged transfer learning, fine-tuning, and data augmentation techniques.
- Hybrid techniques were used that combined histogram of oriented gradient features as well as deep learning features from a residual network to achieve better performance.
- Comparisons between the performance and architectural complexity were carried on for each model to assess how they performed against each other.
- Overall, InceptionV3 set the performance benchmark, achieved perfect performance (100% accuracy, 0 errors), benefiting from aggressive augmentation, strong regularization, and conservative fine-tuning. while DenseNet121 offered the best balance between system resource utilization and performance, making it most suitable for deployment.

FUTURE WORK

- Development of a lightweight deep learning model that can classify multiple classes and can be deployed on edge devices can be considered as future work.

Data Availability:

- The acoustic dataset used for this study by author **Al-Emadi, S.**, can be accessed on Github in the Drone Audio Dataset repository at <https://github.com/saraalemadi/DroneAudioDataset>.
- The image dataset used for this study is by author **Simarpreet Singh**. It can be accessed on Kaggle datasets at <https://www.kaggle.com/datasets/cybersimar08/drone-detection>.

REFERENCES

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